

Project Plan with Gantt Chart

Software Engineering - SVNC.00.308

Project: StayEase Hotel Reservation System

Type: Hotel Reservation

Team: Cat Bacillus (Kyrylo Pryiomyshev, Danylo Chernov, Daniil Ustuhov)

Date: February 20, 2026

Version: 1.0

1. Project Timeline Overview

Project Start Date: February 16, 2026

Target Completion Date: April 30, 2026

Total Duration: 11 weeks / 74 days

2. Work Breakdown Structure (WBS)

Phase 1: Requirements (Week 1-2, 10 days)

- Task 1.1: Stakeholder interviews with hotel manager
- Task 1.2: Requirements documentation
- Task 1.3: Requirements review and approval

Deliverable: Requirements Specification Document

Milestone: Phase Gate 1 - Requirements Approved

Phase 2: Design (Week 3-4, 10 days)

- Task 2.1: System architecture design
- Task 2.2: Database schema design
- Task 2.3: Component interface and API design
- Task 2.4: UI mockup creation
- Task 2.5: Design review and approval

Deliverable: Design Documentation

Milestone: Phase Gate 2 - Design Approved

Phase 3: Implementation (Week 5-8, 21 days)

- Task 3.1: Set up development environment and project scaffold
- Task 3.2: Implement database layer (schema, migrations, seed data)
- Task 3.3: Implement backend API (authentication, rooms, reservations)
- Task 3.4: Implement frontend UI (guest portal, staff dashboard)
- Task 3.5: Component integration and API wiring
- Task 3.6: Code review and refactoring

Deliverable: Working software (alpha version)

Milestone: Phase Gate 3 - Implementation Complete

Phase 4: Testing (Week 9-10, 14 days)

- Task 4.1: Unit testing
- Task 4.2: Integration testing
- Task 4.3: System testing (end-to-end workflows)
- Task 4.4: User Acceptance Testing (UAT) with hotel manager
- Task 4.5: Bug fixes and regression testing

Deliverable: Test reports, validated system

Milestone: Phase Gate 4 - Testing Complete

Phase 5: Deployment (Week 11, 7 days)

- Task 5.1: Production environment setup
- Task 5.2: Data migration (room inventory, staff accounts)
- Task 5.3: System deployment
- Task 5.4: Staff training
- Task 5.5: Go-live

Deliverable: Deployed system, trained staff

Milestone: Phase Gate 5 - Deployment Complete

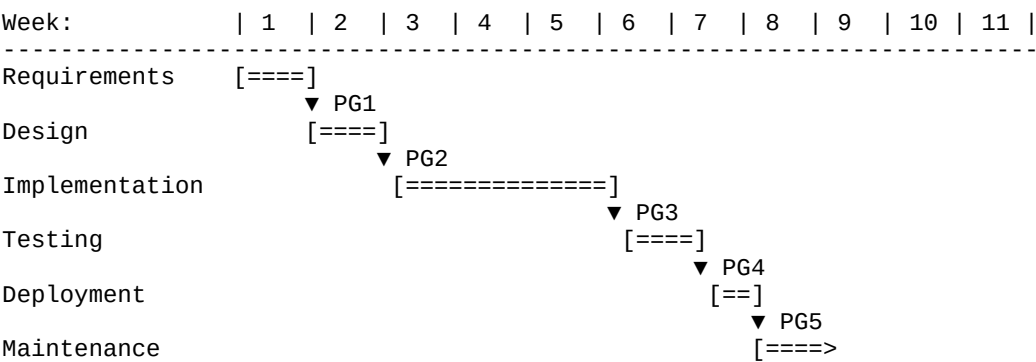
Phase 6: Maintenance (Ongoing)

- Task 6.1: Monitor system performance and uptime
- Task 6.2: Bug fixes and patches
- Task 6.3: User support and feedback collection

Deliverable: Stable, maintained system

3. Gantt Chart

The following text-based Gantt chart visualizes the project timeline across 11 weeks:



4. Detailed Task Table

Task ID	Task Name	Duration	Start	End	Depends	Assigned To
1.1	Stakeholder interviews	3 days	Feb 16	Feb 18	None	Kyrylo
1.2	Requirements docs	5 days	Feb 19	Feb 23	1.1	All team
1.3	Requirements	2 days	Feb 24	Feb 25	1.2	Stakeholde

	review					r
PG1	Phase Gate 1	-	Feb 25	Feb 25	1.3	-
2.1	Architecture design	3 days	Feb 26	Feb 28	PG1	Danylo
2.2	Database schema	3 days	Mar 1	Mar 3	2.1	Daniil
2.3	API/component design	3 days	Mar 1	Mar 3	2.1	Kyrylo
2.4	UI mockups	2 days	Mar 4	Mar 5	2.1	Danylo
2.5	Design review	2 days	Mar 6	Mar 7	2.2-2.4	Stakeholder
PG2	Phase Gate 2	-	Mar 7	Mar 7	2.5	-
3.1	Dev env setup	2 days	Mar 8	Mar 9	PG2	Kyrylo
3.2	Database impl	5 days	Mar 10	Mar 14	3.1	Daniil
3.3	Backend API impl	12 days	Mar 10	Mar 25	3.1	Kyrylo
3.4	Frontend UI impl	12 days	Mar 10	Mar 25	3.1	Danylo
3.5	Integration	5 days	Mar 26	Apr 1	3.2-3.4	All team
3.6	Code review	2 days	Apr 2	Apr 3	3.5	Kyrylo
PG3	Phase Gate 3	-	Apr 3	Apr 3	3.6	-
4.1	Unit testing	4 days	Apr 4	Apr 7	PG3	All team
4.2	Integration testing	3 days	Apr 8	Apr 10	4.1	Daniil
4.3	System testing	3 days	Apr 11	Apr 13	4.2	Daniil
4.4	UAT	3 days	Apr 14	Apr 16	4.3	Stakeholder
4.5	Bug fixes	3 days	Apr 17	Apr 19	4.4	All team
PG4	Phase Gate 4	-	Apr 19	Apr 19	4.5	-
5.1	Prod env setup	2 days	Apr 20	Apr 21	PG4	Kyrylo
5.2	Data migration	2 days	Apr 22	Apr 23	5.1	Daniil
5.3	Deployment	1 day	Apr 24	Apr 24	5.2	Kyrylo
5.4	Staff training	3 days	Apr 25	Apr 27	5.3	All team
5.5	Go-live	1 day	Apr 28	Apr 28	5.4	All team
PG5	Phase Gate 5	-	Apr 28	Apr 28	5.5	-

5. Key Milestones

Milestone	Description	Target Date
Phase Gate 1	Requirements approved	Feb 25, 2026

Phase Gate 2	Design approved	Mar 7, 2026
Phase Gate 3	Implementation complete	Apr 3, 2026
Phase Gate 4	Testing complete	Apr 19, 2026
Phase Gate 5	Go-live, deployment complete	Apr 28, 2026

6. Resource Allocation

Team Member	Role	Primary Responsibilities	Est. Hours
Kyrylo Pryiomyshev	PM / Backend Dev	Planning, coordination, backend API, deployment	130 hrs
Danylo Chernov	Design Lead / Frontend Dev	Architecture, UI/UX design, frontend implementation	110 hrs
Daniil Ustihov	QA Lead / DB Developer	Database, test planning, testing, bug tracking	100 hrs

Total Estimated Effort: 340 person-hours (~8.5 person-weeks)

7. Risks and Mitigation

Risk	Probability	Impact	Mitigation Strategy
Requirements change mid-project	Medium	High	Formal change control process; assess impact before accepting
Team member unavailable	Low	High	Cross-train all members; document decisions and knowledge
Integration issues between frontend and backend	Medium	Medium	Agree on API contracts early; frequent integration testing
UAT reveals major usability issues	Low	High	Demo to stakeholder during development; early feedback loops
Date/timezone handling bugs in reservations	Medium	High	Use ISO 8601 dates; thorough edge case testing for overlapping bookings
Deployment delays	Low	Medium	Prepare production environment early; rehearse deployment process

8. Sign-Off

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Reviewed By: Elena Petrova (Hotel Manager) / Nicolai Morozov (Instructor)

Approval Date: February 23, 2026

Project Manager Signature: _____ Date: _____